

Heart Disease Prediction Project

1. Introduction

This project focuses on predicting the presence of heart disease using clinical and medical attributes. Machine learning techniques are applied to analyze patient data and identify key risk factors. Heart disease is one of the leading causes of death worldwide, making early detection and prevention critically important in modern healthcare systems. Conditions such as coronary artery disease, heart failure, and arrhythmias often develop gradually and may not show clear symptoms in early stages. Therefore, identifying individuals at risk through data-driven approaches has become an essential area of research.

With the rapid growth of healthcare data and advancements in technology, predictive analytics and machine learning have emerged as powerful tools for disease detection. These techniques enable the analysis of large volumes of patient data, including clinical measurements, lifestyle factors, and diagnostic test results, to uncover hidden patterns and relationships that may not be easily identified through traditional methods.

In this project, a heart disease prediction model is developed using key medical attributes such as age, sex, chest pain type, blood pressure, cholesterol levels, and other physiological indicators. By applying machine learning algorithms, the model aims to classify whether a patient is likely to have heart disease or not. This not only helps in early diagnosis but also supports healthcare professionals in making informed clinical decisions.

Furthermore, the use of feature importance analysis allows us to identify the most significant risk factors contributing to heart disease. This insight can assist in preventive strategies and personalized treatment plans. While such models provide valuable support, they are intended to complement—not replace—medical expertise.

Overall, this project demonstrates how data science and predictive modeling can contribute to improving healthcare outcomes by enabling early detection, reducing risks, and enhancing decision-making processes in clinical environments.

2. Objective

To build a predictive model that can classify whether a patient is likely to have heart disease based on medical features.

- To develop a predictive model that can accurately identify the presence of heart disease using patient medical data.
- To analyze important clinical features such as age, sex, chest pain type, blood pressure, cholesterol levels, and heart rate in relation to heart disease risk.
- To apply machine learning algorithms (such as Random Forest) for classification and prediction of heart disease.
- To preprocess the dataset by handling missing values, normalizing data, and preparing it for model training.
- To evaluate the performance of the model using metrics like accuracy, confusion matrix, precision, recall, and F1-score.
- To identify the most significant risk factors influencing heart disease through feature importance analysis.
- To provide a data-driven approach that can assist healthcare professionals in early detection and better decision-making.
- To ensure ethical use of healthcare data by maintaining patient privacy and data confidentiality.

3. Description of Variables

Age: Age of the patient in years.

Sex: Gender of the patient (1 = male, 0 = female).

Chest Pain Type (cp): Type of chest pain experienced (4 categories).

Resting Blood Pressure (trestbps): Blood pressure measured at rest (mm Hg).

Serum Cholesterol (chol): Cholesterol level in mg/dl.

Fasting Blood Sugar (fbs): Whether fasting blood sugar > 120 mg/dl (1 = true, 0 = false).

Resting ECG (restecg): Electrocardiographic results (0,1,2).

Maximum Heart Rate (thalach): Maximum heart rate achieved.

Exercise Induced Angina (exang): Presence of angina due to exercise (1 = yes, 0 = no).

Oldpeak: ST depression induced by exercise relative to rest.

Slope: Slope of peak exercise ST segment.

CA: Number of major vessels colored by fluoroscopy (0–3).

Thal: 0 = normal; 1 = fixed defect; 2 = reversible defect.

4. Methodology

This project follows a systematic approach to develop a heart disease prediction model using machine learning techniques. The methodology consists of the following steps:

1. Data Collection

The dataset used for this study is obtained from a reliable source such as the UCI Machine Learning Repository or Kaggle. It contains medical attributes of patients including age, sex, chest pain type, blood pressure, cholesterol level, and other clinical measurements.

2. Data Preprocessing

Before applying machine learning algorithms, the dataset is cleaned and prepared:

- Missing values are identified and handled using appropriate techniques such as mean imputation.
- Irrelevant or inconsistent data is removed.
- Categorical variables are converted into numerical format if required.
- Feature scaling (normalization/standardization) is applied to ensure uniformity in data.

3. Feature Selection

Important features influencing heart disease are selected based on domain knowledge and statistical relevance. This step helps in improving model performance and reducing complexity.

4. Data Splitting

The dataset is divided into two parts:

- **Training Set (80%):** Used to train the model
- **Testing Set (20%):** Used to evaluate model performance

This ensures that the model generalizes well to new, unseen data.

5. Model Building

A machine learning algorithm such as **Random Forest Classifier** is used to build the prediction model. This algorithm is chosen because of its high accuracy, robustness, and ability to handle complex datasets.

6. Model Evaluation

The performance of the model is evaluated using the following metrics:

- **Accuracy:** Measures overall correctness
- **Confusion Matrix:** Shows correct and incorrect predictions
- **Precision & Recall:** Evaluate classification performance
- **F1-Score:** Balance between precision and recall

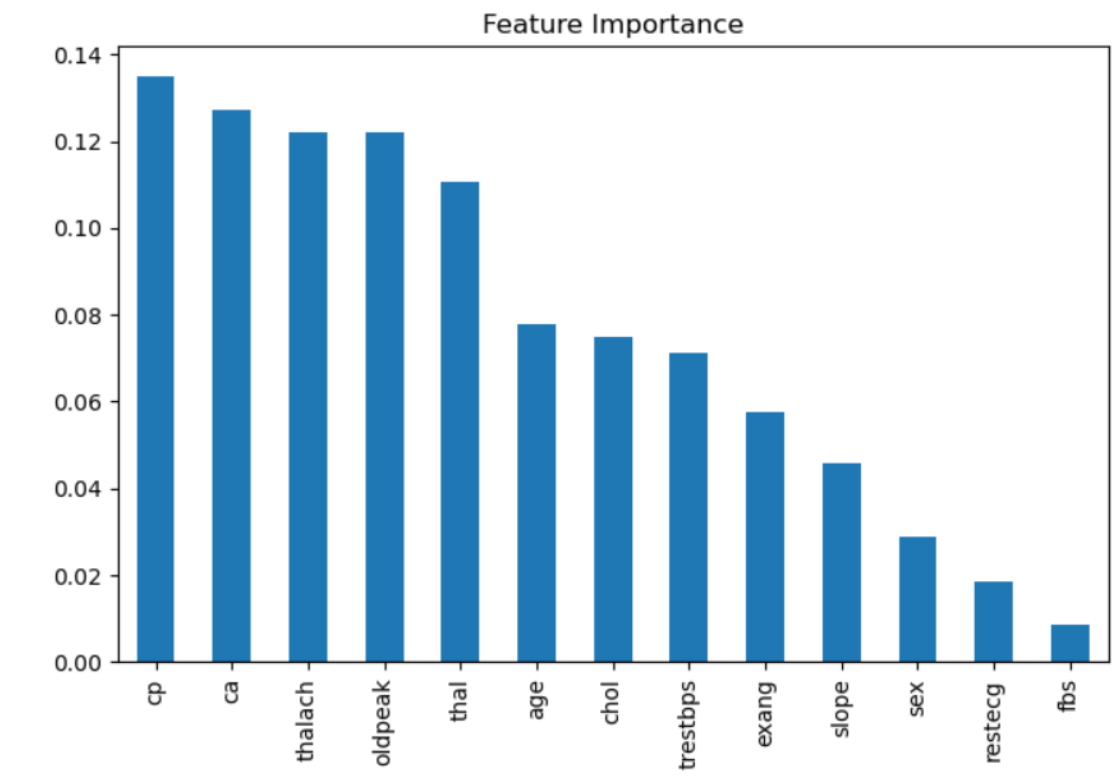
7. Feature Importance Analysis

The model identifies the most influential features contributing to heart disease prediction. This helps in understanding key risk factors such as chest pain type, cholesterol level, and heart rate.

5. Statistical Analysis

Featurre Importance Chart:

Feature importance is a technique used in machine learning to identify which input variables have the greatest influence on the model’s predictions. It helps in understanding how different features, such as age, cholesterol level, or chest pain type, contribute to predicting heart disease. By ranking features based on their importance, we can focus on the most significant risk factors, improve model performance, and make the results more interpretable for better decision-making in healthcare.

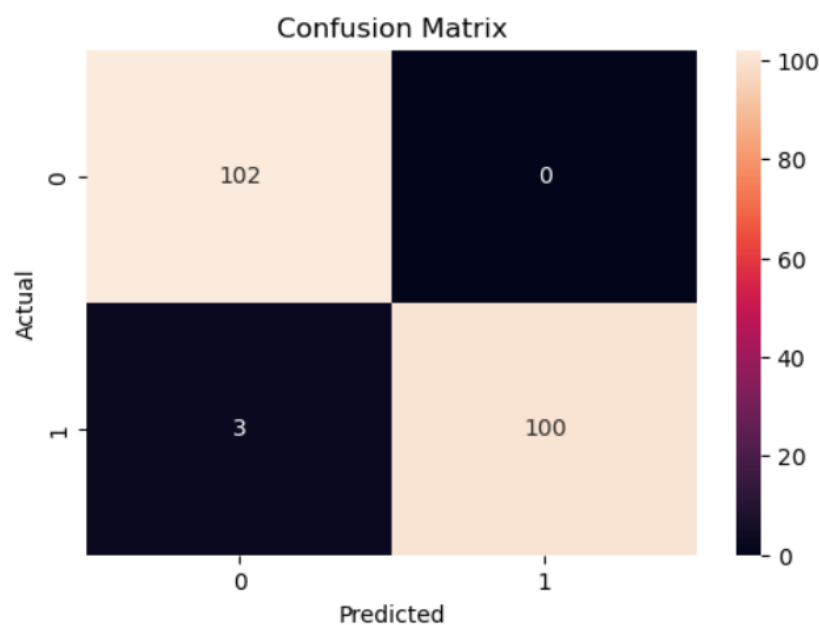


Interpretation:

The feature importance results show that chest pain type (cp) is the most important factor in predicting heart disease, followed by ca and thalach, indicating that heart function and vessel condition are key predictors. Features like oldpeak and thal also have strong influence, while age, cholesterol, and blood pressure have moderate impact. Variables such as sex, resting ECG, and fasting blood sugar contribute very little to the model.

Confusion Matrix

A confusion matrix is a performance evaluation tool used in classification models to compare actual and predicted outcomes. It shows the number of correct and incorrect predictions in terms of true positives, true negatives, false positives, and false negatives. This helps in understanding how well the model is performing and where it is making errors.



Interpretation:

The confusion matrix shows that the model performs very well, with 102 true negatives and 100 true positives, meaning most cases are correctly classified. There are no false positives, indicating the model does not incorrectly predict heart disease for healthy patients, and only 3 false negatives, where actual heart disease cases are missed. Overall, this reflects high accuracy and strong reliability, with very few prediction errors.

Classification Report:

Classification Report:				
	precision	recall	f1-score	support
0	0.97	1.00	0.99	102
1	1.00	0.97	0.99	103
accuracy			0.99	205
macro avg	0.99	0.99	0.99	205
weighted avg	0.99	0.99	0.99	205

Interpretation:

The model achieves an accuracy of approximately 99%, indicating excellent overall performance in predicting heart disease. It correctly classifies almost all cases, with very few misclassifications (only 3 false negatives and no false positives). This means the model is highly reliable, especially in avoiding incorrect positive predictions, though a small number of actual disease cases are missed. Overall, the model demonstrates strong predictive capability and can be considered effective for supporting heart disease detection.

6. Results and Discussion

The developed heart disease prediction model shows excellent performance with an accuracy of approximately 99%, indicating that the model is highly effective in classifying patients correctly. From the confusion matrix, the model achieved 102 true negatives and 100 true positives, which means it correctly identified both healthy individuals and patients with heart disease in most cases. There were no false positives, indicating that the model does not incorrectly label healthy individuals as having heart disease, which is very important in medical diagnosis. However, there were 3 false negatives, meaning a small number of actual heart disease cases were missed.

The feature importance analysis revealed that chest pain type (cp), number of major vessels (ca), and maximum heart rate (thalach) are the most significant factors influencing heart disease prediction. Other variables such as oldpeak and thal also contributed notably, while factors like sex, resting ECG, and fasting blood sugar had minimal impact on the model.

Overall, the results indicate that clinical and exercise-related features play a major role in detecting heart disease, and the model successfully captures these relationships.

7. Conclusion

This project successfully developed a machine learning model for heart disease prediction using clinical data. The model achieved high accuracy (99%) and demonstrated strong ability to correctly classify both diseased and non-diseased cases. The analysis highlighted that heart-related features such as chest pain type, heart rate, and vessel condition are key predictors of heart disease.

Despite its high performance, the model has a small limitation in missing a few positive cases (false negatives), which should be considered in real-world applications. Therefore, this model should be used as a support tool for healthcare professionals rather than a replacement for medical diagnosis.

In conclusion, the study demonstrates that machine learning techniques can effectively assist in early detection of heart disease, improve decision-making, and contribute to better healthcare outcomes when used responsibly and ethically.